

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Functional Requirements:

FR-1: Data Visualization Requirements

- **FR-1.1: Display 12 professional Tableau visualizations including bar charts, line charts, pie charts, heat maps, and scatter plots**
- **FR-1.2: Provide interactive dashboard with dynamic filtering by state, region, and time period**
- **FR-1.3: Enable individual visualization pages with detailed information and insights**
- **FR-1.4: Support 5-scene data story with narrative progression**
- **FR-1.5: Implement responsive visualization containers that adapt to screen sizes**

FR-2: Web Application Requirements

- **FR-2.1: Develop Flask web application with 15+ routes for different pages and functionality**
- **FR-2.2: Create main portfolio page with scenario-based navigation**
- **FR-2.3: Implement three analysis scenarios: Time Patterns, Seasonal Variations, Sector-Specific**
- **FR-2.4: Provide breadcrumb navigation for user orientation**
- **FR-2.5: Support error handling and professional fallback content**

FR-3: Data Management Requirements

- **FR-3.1: Process and analyze 16,599 electricity consumption records**
- **FR-3.2: Support data from 35 Indian states across 5 regions**
- **FR-3.3: Handle monthly data for 2019-2020 time period**
- **FR-3.4: Enable data validation and sanitization processes**
- **FR-3.5: Support data export in multiple formats**

FR-4: User Interface Requirements

- **FR-4.1: Implement responsive design using Bootstrap 5 framework**
- **FR-4.2: Provide color-coded analysis scenarios (blue, orange, green)**
- **FR-4.3: Create professional gradient backgrounds and modern styling**
- **FR-4.4: Support mobile-friendly navigation and touch interactions**
- **FR-4.5: Include hover effects and smooth transitions**

FR-5: Integration Requirements

- **FR-5.1: Integrate Tableau Public visualizations with individual sheet URLs**
- **FR-5.2: Support URL parameter handling for dynamic filtering**
- **FR-5.3: Implement iframe embedding with responsive sizing**
- **FR-5.4: Provide error handling for missing or unavailable visualizations**
- **FR-5.5: Support external API integration for future enhancements**

Non-functional Requirements:

NFR-1: Performance Requirements

- NFR-1.1: Page load time shall be less than 3 seconds for all pages
- NFR-1.2: Visualization loading time shall not exceed 5 seconds
- NFR-1.3: Application shall support 50 concurrent users without performance degradation
- NFR-1.4: Response time for navigation and filtering shall be under 500ms
- NFR-1.5: Mobile performance score shall be 95+ on Google PageSpeed

NFR-2: Usability Requirements

- NFR-2.1: Interface shall be intuitive requiring no technical training for basic use
- NFR-2.2: Any visualization shall be accessible within 3 clicks from main page
- NFR-2.3: Application shall be WCAG 2.1 AA compliant for accessibility
- NFR-2.4: Interface shall support keyboard navigation and screen readers
- NFR-2.5: Touch targets shall be minimum 44px for mobile accessibility

NFR-3: Compatibility Requirements

- NFR-3.1: Application shall work on Chrome, Firefox, Safari, and Edge (latest versions)
- NFR-3.2: Responsive design shall support screen sizes from 320px to 1200px+
- NFR-3.3: Application shall function on iOS Safari and Chrome Mobile
- NFR-3.4: Cross-browser compatibility shall achieve 95% feature parity
- NFR-3.5: Application shall support both desktop and tablet devices

NFR-4: Security Requirements

- NFR-4.1: All user inputs shall be sanitized to prevent XSS attacks
- NFR-4.2: Application shall implement CSRF protection for form submissions
- NFR-4.3: Data transmission shall use HTTPS encryption
- NFR-4.4: Application shall log security events and errors
- NFR-4.5: Sensitive data shall be encrypted at rest

NFR-5: Reliability Requirements

- NFR-5.1: Application uptime shall be 99% during business hours
- NFR-5.2: Error rate shall not exceed 1% of total requests
- NFR-5.3: Application shall handle missing Tableau URLs gracefully
- NFR-5.4: System shall recover from errors without user intervention
- NFR-5.5: Data integrity shall be maintained during all operations

NFR-6: Scalability Requirements

- NFR-6.1: System shall support dataset expansion to 50,000+ records
- NFR-6.2: Architecture shall support addition of new visualization types
- NFR-6.3: Application shall support multiple concurrent analysis scenarios
- NFR-6.4: Database queries shall remain efficient with data growth
- NFR-6.5: System shall support horizontal scaling for increased user load

NFR-7: Maintainability Requirements

- NFR-7.1: Code shall follow PEP 8 Python standards and HTML5 semantic structure
- NFR-7.2: Documentation shall cover 100% of code functionality
- NFR-7.3: System shall use modular architecture for easy maintenance
- NFR-7.4: Configuration shall be externalized for environment flexibility
- NFR-7.5: Code shall have unit test coverage of at least 80%

NFR-8: Documentation Requirements

- **NFR-8.1:** Complete technical documentation shall be provided
- **NFR-8.2:** User installation guide shall be included
- **NFR-8.3:** API documentation shall be comprehensive and current
- **NFR-8.4:** Troubleshooting guide shall cover common issues
- **NFR-8.5:** Documentation shall be updated with each release

Requirements Summary Tables

Table 1: Functional Requirements Priority Matrix

Requirement ID	Priority	Complexity	Development Effort	User Impact
FR-1	Critical	High	High	Very High
FR-2	Critical	Medium	Medium	Very High
FR-3	High	Medium	Medium	High
FR-4	High	Low	Low	High
FR-5	Critical	High	High	Very High

Table 2: Non-functional Requirements Compliance

Requirement Category	Target Metric	Acceptable Range	Testing Method
Performance	<3 seconds	2-4 seconds	Load testing
Usability	WCAG 2.1 AA	90%+ compliance	Accessibility testing
Compatibility	95%+	90-100%	Cross-browser testing
Security	Zero vulnerabilities	0-1 low risk	Security audit
Reliability	99% uptime	95-100%	Monitoring

Table 3: Requirements Traceability Matrix

Business Need	Functional Requirement	Non-functional Requirement	Test Case
Data Visualization	FR-1.1, FR-1.2	NFR-1.1, NFR-2.1	TC-VIZ-001
Web Application	FR-2.1, FR-2.2	NFR-1.2, NFR-3.1	TC-WEB-001
User Experience	FR-4.1, FR-4.2	NFR-2.1, NFR-2.2	TC-UX-001
System Integration	FR-5.1, FR-5.2	NFR-4.1, NFR-5.1	TC-INT-001

Requirements Validation Criteria

Functional Requirements Validation

- All 12 visualizations display correctly with Tableau integration
- Flask application routes respond properly with expected content
- Data processing handles 16,599 records without errors
- Responsive design works across all specified devices
- Error handling provides professional fallback content

Non-functional Requirements Validation

- Performance testing confirms load times under 3 seconds
 - Accessibility testing validates WCAG 2.1 AA compliance
 - Cross-browser testing confirms 95% feature parity
 - Security testing confirms no critical vulnerabilities
 - Reliability testing confirms 99% uptime capability
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Requirements Management

Change Control Process

- Requirements changes documented through formal change requests
- Impact analysis conducted for all proposed changes
- Stakeholder approval required for requirement modifications
- Updated requirements communicated to all team members
- Test cases updated to reflect requirement changes

Requirements Maintenance

- Regular review of requirements for continued relevance
- Documentation updates with each requirement change
- Traceability matrix maintained throughout project lifecycle
- Requirements versioning for change tracking
- Periodic validation against user needs and business objectives

Feedback submitted

Command Awaiting Approval